Hugo Marquerie 17/06/2026

Lem Carac Convergencia Uniforme Compactos Convergencia Localmente Uniforme

Lema 1. *Sea* $\Omega \subset \mathbb{C}$ *y sean* $(f_n)_{n\in\mathbb{N}} \subset \mathcal{C}(\Omega)$ *y* $f \in \mathcal{C}(\Omega)$, *entonces,*

$$f_n \xrightarrow[n\to\infty]{\Omega} f \iff \forall z \in \Omega : \exists r > 0 : \overline{D}(z,r) \Subset \Omega \wedge \lim_{n\to\infty} \sup_{w\in\overline{D}(z,r)} |f_n(w) - f(w)| = 0.$$